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Application of Data Mining Techniques to Developing Classification Model for Glaucoma Type Identification

Abstract

Data mining is also known as Knowledge Discovery in Database (KDD) and is a process which includes extracting the interesting, interpretable and useful information from the raw data. Glaucoma affects through increasing an intraocular pressure (IOP), which is responsible for the glaucomatous optic neuropathy involving the death of retinal ganglion cells and their axons which resulting in blindness. A person, who is responsible to treat glaucoma patients might make an error in glaucoma type identifying and treatment ordering due to subjective decision, knowledge limitation, and visualization through instruments. This results in resource wastage as well as time-consuming. The main aim of this research is to reduce (not removing the problem at all) the biased decision of the ophthalmologist through making an easy, quick and accessible way for glaucoma type identification by developing a better classification model. In this study, data mining techniques are used to discover new knowledge based on the collected dataset. From numerous data mining classification Algorithms, this paper applies naïve Baye, grip, J48, and PART algorithms that are used with two basic test options based on complete and selected features. Based on the experiential analysis performed, the PART algorithm with a test option of 10 fold cross-validation using the selected feature scored the highest accuracy result which is 71.4%.

Keyword: - Machine Learning, Data Mining, Classification, Eye diseases, Glaucoma, Knowledge Base system

1. INTRODUCTION

Glaucoma is a category of eye diseases characterized by progressive optic neuropathy and visual field defect [1] and considered by damage to the optic nerve with corresponding visual field loss. In other words, it is characterized by increased intraocular pressure (IOP), which is responsible for the glaucomatous optic neuropathy involving the death of retinal ganglion cells and their axons [2] which in turn causes damage of the optic nerve, resulting in blindness. In glaucoma, the drainage of aqueous humor through trabeculae is blocked, resulting in increased IOP. When the IOP is above 60 mm Hg, the optic nerve fibers at the optic disk are compressed. Initially it decreases the visual field, which eventually leads to total blindness. Untreated glaucoma leads to permanent damage of the optic nerve and results in blindness [3]. Typically, patients initially lose the mid-periphery of their visual field, while central vision tends to be involved later. The patients become aware of a functional defect when visual field loss impinges upon or involves central vision [4, 5]. Patients with glaucoma may experience difficulty in identifying faces, steering, reading, noticing objects in their peripheral vision, and adapting to different levels of lighting. Moreover, they are also at an enlarged risk of falls and accidents [6]. Fifteen percent of the world's blindness is attributed due to glaucoma and around 600,000 people go blind annually [2]. In 2013, the number of people with glaucoma worldwide was projected to be 64.3 million [7]. This is expected to rise to 76 million in 2020 and 111 million in 2040 [7]. Africa accounts for 15% world's blindness burden due to glaucoma [7]. In Ethiopia, glaucoma is the fifth common cause of blindness which results in an irreversible sight loss for an estimated 62,000 Ethiopians [8, 9]. The increasing prevalence of glaucoma is expected to cause a significant economic burden and poor quality of life [10-12] High glaucoma morbidity among some African communities may be attributed to low trained professionals, low awareness, and under-utilization of eye care service [13-15]. It has been estimated that half of the glaucoma patients are already blind in at least one eye in presentation in Africa [16]. Data itself is nothing, but to process it, is very useful and motivating. The way of identifying glaucoma type is so difficult and handle through data mining technology by discovering new and hidden knowledge.

2. LITERATURE REVIEW

Maila C et.al. [17] conducted paper entitled automatic glaucoma detection based on optic disc segmentation and texture feature extraction with the major aim of developing an automatic detection method of glaucoma in the retinal image to check whether or not the eye become glaucomatous via classifying the retinal image. To achieve the final objective, the methods used by the researchers were the acquisition of image databases, optic disk segmentation, and texture feature extraction in different color modes.

The classifier algorithms adopted by their paper were multiplying perceptron, random committee, random forest, and radial basis function. Finally, the researchers obtained the highest accuracy result of 93.03 % by the MLP. Kinjan Chauhan and Ravi Gulati [18] have conducted paper adopting a data mining approach with the major aim of detecting and diagnosing glaucoma disease using the Perimetry and Optical Coherence Tomography (OCT) images and predicting the diagnosis for progression of glaucoma. Additional to the image procession techniques they have used for feature extraction, some of the data mining algorithms they adopted were neural network, decision tree, logistic regression, machine learning classifier (MLC), linear discriminant analysis, and support vector machine (SVM). Finally, the SVM and Naïve Bayes algorithms achieved the highest accuracy value, and accordingly, the model is developed for detecting glaucoma based on a decision of whether the glaucoma ailment is absent or present in the eye. The paper was done by Sadaf Malik et al. [19] aimed at developing a classification model for classifying the different types of eye diseases by following a data-driven approach and by adopting machine learning and data mining techniques such as decision tree, naive Bayes, random forest and neural network from which the random forest algorithm achieved the highest accuracy result (86.36%). Finally, the score point of 86.36% has been registered via the algorithm of random forest. The paper was focused on classifying diseases by considering glaucoma as one class label. In [20] paper entitled "classifying chief complaint in eye diseases using data mining techniques", has been conducted to discover new hidden knowledge for classifying the different types of eye disease using the intelligent capability of data mining technologies like MLP, ANN and naïve baye from which the last two achieved the highest accuracy. Like all the literature reviewed, this paper also uses glaucoma as one of the class labels, and the last target is only classified the eye disease as one of the types. In [21] paper entitled "Data Mining Techniques for Diagnostic Support of Glaucoma using Stratus OCT and Perimetric data for the identification of glaucoma from other diseases in the case of retinal image processing. The Parameters obtained from the Perimetry and Stratus Optic Coherence Test (OCT) have been fed to each technique to find out their performance in terms of accuracy, sensitivity, and specificity by using the data mining technique. The data mining algorithms used for diagnostic classification were Decision Tree, Linear Regression, and Support Vector Machine (SVM). Among those algorithms, the Decision Tree and Linear Regression Model performs much better than other algorithms for the diagnosis of Glaucoma by achieving the highest accuracy of 92.56% and 70.25% respectively. The specificity of Linear Regression and SVM is 97.56% and 96.34% respectively. A paper done in [22] applied a machine learning techniques such as fuzzy logic, decision tree based on ID3 algorithms, support vector machine, and k-nearest neighbor. Among those techniques, the last three achieved the highest accuracy result of 85, 80, and 86%. In [23] a paper has been done for glaucoma detection and prediction using data mining classification algorithms namely, KNN and Naïve Baye. The final analysis was decided on 65% accuracy result scored through the Naïve Baye algorithm. The basic difference with this work is that the related paper is focused on detect glaucoma disease.

3. METHODOLOGY

The paper methodology includes research design and implementation tools. It is characterized by utilizing those apparatus to accumulate significant data in a particular research study. To conduct this study, this researcher used interviews, document as well as data analysis mechanisms. Mainly, this paper is adopted a design science approach with the reasonable point of focusing around on making data that is used to deal with the issue looked in the area, used for creating new artifacts from the given dataset and Design science research accomplishes knowledge and understanding of a problem through the way of establishing a designed artifact and so on. This study is conducted through using the data mining, classification algorithms and other tools like rapid miner for data preprocessing and WEKA tool for knowledge generation and follow the KDD process model.

Data preprocessing

Data preprocessing is a major task of data mining technique that is used to form data analysis and transform the raw data into a useful and efficient format. This data preprocessing is mainly concerned with the task of missing value handling with a method of ignoring or fill the values, though different methods, data cleansing, data transformation with normalization and discretization. In this paper, the collected dataset has some missing values, imbalance classes, and needs feature selection. The missing value was handled by replacing the maximum frequent value in the recorded dataset. In detail discription, the missing value of each dataset is a full field or handled through replacing by maximum frequent value in the dataset. Whereas feature selection is operated through an information gain value. The information gain value of each attribute was calculated through calculating the entropy of classes (four classes namely, primary open-angle glaucoma, primary angle-closure glaucoma, secondary glaucoma, and congenital glaucoma) and calculating the gain value of each attribute and subtracting it from the entropy and put them orderly from highest to lowest.

The other remaining data cleaning procedure is data discretization and is apply on an age parameter. The class imbalance problem has been improved by SMOTE analysis. The dataset were contain 4166 instances after SMOTE with 18 attributes and the feature were selected via their information gain value.

Table 1. 1 Information Gain value of attributes

No	Attribute name	Information Gain Value
1.	'Headache'	0.127
2.	'age '	0.091
3.	'Nausea'	0.074
4.	'sudden eye pain'	0.072
5.	'frequent eye trauma'	0.066
6.	'Sudden sight loss'	0.058
7.	'long sight effect'	0.053
8.	'Itchy'	0.047
9.	'Vomiting'	0.040
10.	'Tear percolate'	0.036
11.	'Living Area'	0.035
12.	'Eye domineer '	0.025
13.	'Eye hoodness'	0.023
14.	'cloudy cornea'	0.011
15.	'sex '	0.006
16.	'Eye casing distent '	0.006
17.	'Blurred vision'	0.005
18.	'Light sensitivity'	0.003

4. THE PROPOSED MODEL

The raw data are collected from the selected organization. I.e. BMH. But the data existed in the manual case and the researcher organized these manual cards to Microsoft word excel understanding format. After gathering the data, it needs pre-processing activities like missing value handling, feature selection, data discretization, and other tasks. So, after completing the pre-processing task of the dataset, the model is developed after empirical analysis of four different classification algorithms, namely Naïve Baye, Jrip, J48, and the PART algorithm was used with two basic test options namely, 10 fold cross-validation with its default value and percentage split of the dataset with 80/20. The reason for using only four classification algorithms is that after trying other classification algorithms those four algorithms have better accuracy than others, are easy to understand and interpret the model outcome and classify hidden patterns. The attribute used for model development was either in the whole attribute or selected attribute. From those four algorithms, one algorithm was selected (highest accuracy result) by comparing others based on two test options using either in the whole or selected attribute. Finally, one model is selected (highest accuracy result algorithms based on either 10 fold cross-validation or 80/20 percentage split, using either in the whole or selected attribute).

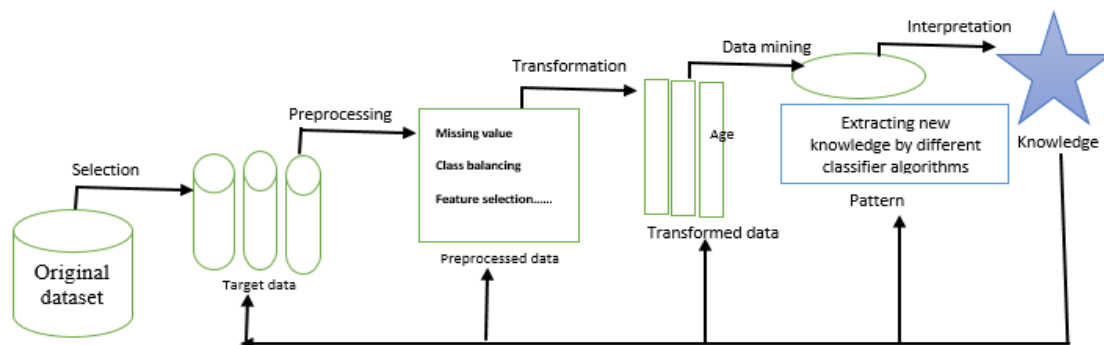


Figure 1. 1 Respected with KDD Process

5. EXPERIMENTAL SET UP

Primarily, to conduct the experiment there must be understanding features with their information gain value. So, the ordering impact of attributes for classifying of glaucoma ailment is conducted by the WEKA tool after the data were preprocessed through rapid miner. The dataset of glaucoma recorded from problem selected area (Borumbeda hospital) was converted into an ARFF for the WEKA tool understandable format.

Test options

For experimenting, the researcher selects two basic test options with coordinating different (four) types of algorithms. The dataset should be converted to ARRF (Attribute Relation File Format) file to understand the format for analyzing the software.

K- Fold cross-validation: -Cross-validation is a resampling process that has the functionality of evaluating machine learning models on a data section.

The procedure has a single parameter called k that refers to the number of groups that a given data to be split. It is making a fold of a dataset into some folds or cataloging the data into k folds. In this paper, the researcher uses 10fold cross-validation with its default value.

Percentage split: -This type of test option is applying a partitioning method of a dataset into training and test set in percentage. In the case of this paper, the researcher uses 80/20 splitting test option means that 80% of the dataset is used for training purposes and the remaining 20 % is used for testing determination.

Experimental Conducted Algorithms and processing action

Naive Bayes algorithm: - it is a classification algorithm that is based on Bayes theorem with strong and naïve dependence free assumptions and is the simplest probabilistic classifiers. The Naïve Bayes classifiers are used for any classification in machine learning based on the conditional possibility or probabilities of the features attributed to a class, in which the features are selected by feature selection methods [24].

PART algorithm: - PART (Projective Adaptive Resonance Theory) is simply a separate and conquer rule learner and producing sets of decision lists rules. A new data are compared to each rule in the list in turn, and the item is assigned the class of the first matching rule. PART builds a decision tree in each iteration and makes the best leaf into a rule [25].

J48 Algorithm: - is an extension of ID3. An added features of J48 are secretarial for missing values, decision trees clipping, continuous attribute value ranges, derivation of rules, etc.

The WEKA tool provides several options associated with tree trimming. In the case of potential overfitting, trimming can be used as a tool for préising. In other algorithms the classification is performed recursively till every single leaf is pure, that is the classification of the data should be as perfect as possible. This algorithm generates the rules from which particular identity of that data is generated. The objective is gradually generalization of a decision tree until it gains the balance of elasticity and accuracy [26].

Jrip Algorithm: - Jrip is Repeated Incremental Pruning to Produce Error Reduction (RIPPER) which is the most popular algorithm and uses sequential covering algorithms for creating ordered rule lists.

Classes are examined in growing size and an initial rule for the class is produced using incrementally reduced error Jrip (RIPPER) proceeds by treating all the examples of a particular decision on the training data as a class and finding a set of rules that cover all the members of that class. Thereafter, it proceeds to the next class and does the same, repeating this until all classes have been covered [27].

6. EXIPERIMENT AND RESULT

There are around sixteen experiments were conducted using four basic classification algorithms namely Jrip, J48, PART and Naïve Baye based on two basic test options 10 fold cross validation and percentage split (80/20) with comparing the experimental result via selected or whole parameters usage. The performance metrics of those experiments are accuracy, TPR, FPR, and F-score.

Table 1. 2 Total experimental result

Test mode with Attribute	Classifier algorithms				
	Performance metrics	Jrip	Naive aye	J48	PART
10 fold cross-validation/ whole attribute	Accuracy	63.5%	50.3%	68.5%	68.3%
	Precision	65%	50%	68%	68%
	Recall	63%	50%	68%	68%
	TPR	65%	50%	65%	68%
	FPR	12%	16%	10%	10%
	F-score	63%	49%	68%	68%
	ROC area	79%	74%	83%	80%
Percentage Split 80/20/ whole attribute	Accuracy	62.6%	47.6%	68%	66%
	Precision	64%	49%	68%	66%
	Recall	62%	47%	68%	66%
	TPR	52%	47%	68%	66%
	FPR	12%	17%	10%	11%
	F-score	62%	46%	68%	66%
	ROC area	83%	77%	84%	82%
10 fold cross-validation/ selected attribute	Accuracy	67.4%	56.6%	69%	71.4%
	Precision	68%	57%	70%	72%
	Recall	67%	56%	65%	72%
	TPR	67%	56%	70%	69%
	FPR	10%	14%	10%	9%
	F-score	67%	55%	70%	72%
	ROC area	84%	81%	86%	87%
Percentage Split 80/20/ selected attribute	Accuracy	67.2%	57.6%	67.5%	67.4%
	Precision	68%	59%	68%	68%
	Recall	67%	57%	67%	67%
	TPR	67%	57%	67%	67%
	FPR	10%	14%	10%	10%
	F-score	67%	57%	67%	67%
	ROC area	83%	83%	84%	85%

7. CONCLUSION AND FINDING

Finally, based on the dataset collected from health organization (BMH) a new knowledge has been discovered and high accuracy has achieved 71.4% by PART algorithm in order to identify types of glaucoma among available treated type of glaucoma in Borumeda hospital integrated with knowledge extracted from expert and existence document. The final model has get an acceptance of 82.2% by final user.it indicates it will good, if it apply at the organization.

8. FUTURE WORK

The research was conducted based on the researcher's belief and direction. Furtherly, other researchers will modify and update this research through making the research as country based by taking dataset from different health organization, only use PART algorithm based on final accuracy, but may combination of algorithms will be best accuracy result (ensemble) and The research follow KDD process rather use hybrid for future may good as much.

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